Elimination with Matrices

Solve $Ax = b$ the way every computer does: eliminate down to an upper triangular U, then back-substitute. And see every step as a matrix.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **The idea of elimination**
   Turn the system into a triangle you can solve from the bottom up. Pivots and multipliers.
2. **2 × 2 by elimination**
   The two-fund portfolio in one step, and what elimination does to the row picture.
3. **3 × 3, step by step**
   The three-fund portfolio: augmented matrix, three elimination steps, the upper triangular U.
4. **Back substitution**
   Solve $Ux = c$ from the last equation up.
5. **When elimination breaks down**
   A 0 in the pivot position: fixed by a row exchange, or a sign the matrix is singular.
6. **Elimination as matrices**
   Each step is a multiplication by an elimination matrix E. Row times matrix combines rows.
7. **Row exchanges: permutations**
   The matrix P that swaps rows, and why $PA \neq AP$.
8. **Undoing a step: inverses**
   Every step can be reversed, which is why elimination gives the right answer.

METHODS IN THIS LESSON: Gaussian elimination · Augmented matrix · Back substitution · Row exchange (pivoting) · Elimination matrices · Permutation matrix · Inverse of a step

PART 2 Prerequisites

From Lesson 1: The Geometry of Linear Equations. Click any link to review it, then come back.

- Matrix form $Ax = b$: rows of A are rules, columns are funds.

- Matrix × vector, two ways: Ax is a combination of the columns of A.
- The row picture: each equation is a line (or plane).
- The three-fund portfolio (3 × 3): the system we solve here.
- Singular matrices: columns that add no new direction.

PART 3 Explanation

THE RUNNING EXAMPLE: THE SAME PORTFOLIO, SOLVED PROPERLY

In Lesson 1 we split **₹10 lakh** between funds to hit a budget, a return and a market exposure. We found the answers by drawing pictures and by spotting tricks. That doesn't scale: with 50 funds and 50 rules nobody can draw or guess.

This lesson gives the method every computer uses: **elimination**. We apply it to the same two systems:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}}_{\text{Nifty + bonds}} \qquad \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix}}_{\text{Nifty + bonds + midcap}}$$

Rows: budget (₹ lakh), return (₹ thousand), market exposure (₹ lakh). Unknowns: ₹ lakh in each fund.

1. The idea of elimination

METHOD · GAUSSIAN ELIMINATION (FORWARD ELIMINATION)

Subtract multiples of each pivot row to create zeros below the pivot, column by column, until the matrix is upper triangular. Named after Carl Friedrich Gauss (1777–1855), though the method is far older: it appears in China around 2,000 years ago, in *The Nine Chapters on the Mathematical Art*. It is the algorithm inside every equation solver.

Some systems are easy. If the last equation has only one unknown, you solve it at once, then work upwards:

$$\begin{aligned} x + y + z &= 10 \\ -5y + 3z &= -9 \\ -0.4z &= -0.8 \end{aligned} \qquad \Rightarrow \qquad z = 2,\ y = 3,\ x = 5.$$

Its matrix is upper triangular: zeros everywhere below the diagonal. **Elimination** turns any (good) system into one like this, without changing its solution.

> **ELIMINATION IN ONE SENTENCE**
> Use the first equation to remove x from every equation below it; then use the new second equation to remove y from every equation below it; and so on, until the matrix is upper triangular. Strang: "it's the way every software package solves equations."

Two words do all the work:

- The pivot is the diagonal entry we use to clear the entries below it.
- The multiplier is how many times we subtract the pivot row: **multiplier = (entry to clear) ÷ (pivot)**. The multiplier for row i, column j is written ℓ_{ij} ("ell").

> **WHY SUBTRACTING EQUATIONS IS ALLOWED**
> If two statements are true, their difference is true too. Subtracting 12 × (budget rule) from the return rule gives a new rule that every correct portfolio still satisfies, and we can undo it by adding back (topic 8). So the solution never changes.

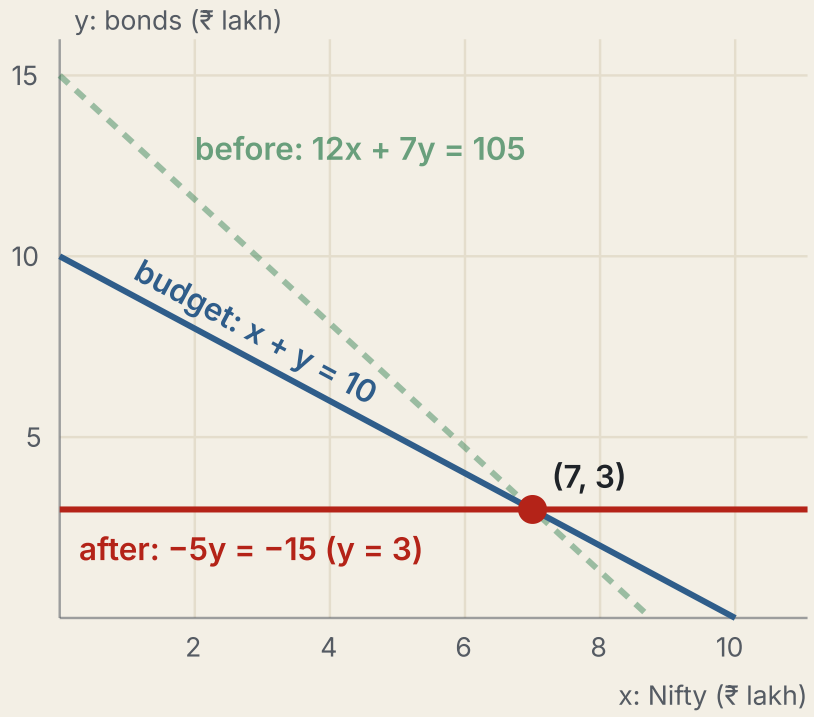
2. 2 × 2 by elimination

The two-fund portfolio: $x + y = 10$ (budget) and $12x + 7y = 105$ (return).

1. **Pivot:** the 1 in the top-left corner.
2. **Multiplier:** the entry to clear is 12, so $\ell_{21} = 12/1 = 12$.
3. **Subtract** 12 × row 1 from row 2, on both sides: $(12x + 7y) - 12(x + y) = 105 - 12(10)$, which is $-5y = -15$.

$$\left[\begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 12 & 7 & 105 \end{array}\right] \xrightarrow{R_2 - 12R_1} \left[\begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 0 & \boxed{-5} & -15 \end{array}\right]$$

Now back-substitute: $-5y = -15$ gives $y = 3$, then $x + 3 = 10$ gives $x = 7$. ₹7 lakh Nifty, ₹3 lakh bonds, the same answer as Lesson 1. The pivots are 1 and -5.



What elimination does to the row picture. The return line (dashed) is replaced by the horizontal line $y = 3$. Different line, **same crossing point** $(7, 3)$: the solution didn't change, it just became easy to read.

▼ **Check yourself (MIT problem 2.1): eliminate $2x + 3y = 5, 6x + 15y = 12$. What are the multiplier and the pivots?**

$\ell_{21} = 6/2 = 3$. Row 2 − 3 × row 1 gives $6y = -3$. Pivots 2 and 6. Back-substitute: $y = -\frac{1}{2}$, then $2x = 5 + \frac{3}{2}$, so $x = \frac{13}{4}$.

3. 3 × 3, step by step

METHOD · AUGMENTED MATRIX $[A \mid B]$

Write b as an extra column next to A, so every row operation is applied to both sides of the equations at once.

Write the coefficients and the right-hand side side by side. Tacking b on as an extra column gives the augmented matrix $[A \mid b]$, so whatever we do to an equation's left side we also do to its right side.

$$[A \mid b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 15 & 111 \\ 1 & 0 & 1.2 & 7.4 \end{array}\right] \quad \begin{array}{c} \text{budget} \\ \text{return} \\ \text{exposure} \end{array}$$

Column 1: clear everything below the first pivot

The first pivot is the 1 in the top-left corner.

- **Step (2, 1):** $\ell_{21} = 12/1 = 12$. Row 2 − 12 × row 1: $(12, 7, 15, 111) - (12, 12, 12, 120) = (0, -5, 3, -9)$.
- **Step (3, 1):** $\ell_{31} = 1/1 = 1$. Row 3 − 1 × row 1: $(1, 0, 1.2, 7.4) - (1, 1, 1, 10) = (0, -1, 0.2, -2.6)$.

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & -5 & 3 & -9 \\ 0 & -1 & 0.2 & -2.6 \end{array}\right]$$

The Nifty amount x is now gone from rules 2 and 3. They form a smaller 2×2 system in y and z, and we simply repeat.

Column 2: clear below the second pivot

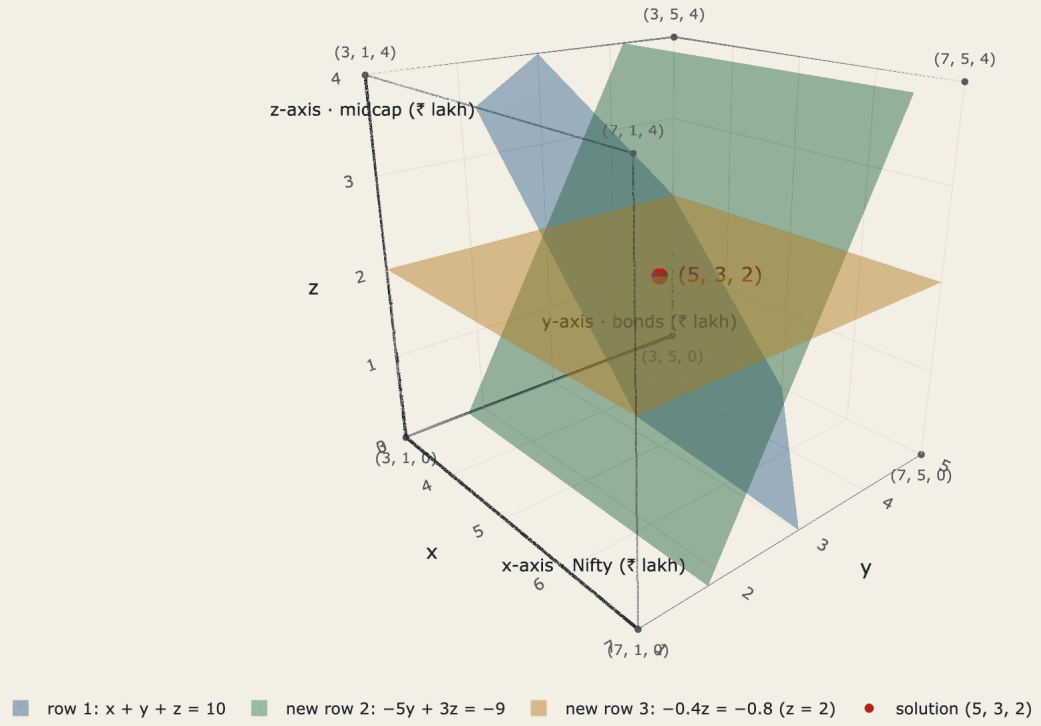
The second pivot is the -5 that has appeared in row 2, column 2.

- **Step (3, 2):** $\ell_{32} = (-1)/(-5) = 0.2$. Row 3 − 0.2 × row 2: $(0, -1, 0.2, -2.6) - (0, -1, 0.6, -1.8) = (0, 0, -0.4, -0.8)$.

$$[U \mid c] = \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-5} & 3 & -9 \\ 0 & 0 & \boxed{-0.4} & -0.8 \end{array}\right]$$

> **RESULT OF ELIMINATION**
>
> A has become the upper triangular matrix U, and b has become $c = (10, -9, -0.8)$. The new system $Ux = c$ has **exactly the same solution** as $Ax = b$. The **pivots** $1, -5, -0.4$ sit on the diagonal of U.

A preview Strang drops in passing: multiply the pivots and you get the *determinant* of A. Here $1 \times (-5) \times (-0.4) = 2$. Determinants come much later.



The row picture, before and after elimination (use the buttons; drag to rotate). Elimination swaps the return and exposure planes for new ones, ending with the flat plane $z = 2$. All three still meet at the same point $(5, 3, 2)$.

4. Back substitution

METHOD · BACK SUBSTITUTION

Solve an upper triangular system $Ux = c$ from the **last** equation up. Each equation brings in exactly one new unknown.

$Ux = c$ is triangular, so solve it **from the bottom up**. Each equation has only one new unknown:

1. **Row 3:** $-0.4z = -0.8$, so $z = 2$.
2. **Row 2:** $-5y + 3(2) = -9$, so $-5y = -15$ and $y = 3$.
3. **Row 1:** $x + 3 + 2 = 10$, so $x = 5$.

₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap, the answer Lesson 1 found by substitution. Check it in the **original** rules: $5 + 3 + 2 = 10$ ✓, $60 + 21 + 30 = 111$ ✓, $5 + 0 + 2.4 = 7.4$ ✓.

THE WHOLE METHOD

Forward elimination ($A \to U$, $b \to c$), then **back substitution** ($Ux = c$, bottom row first). That is all a computer does to solve $Ax = b$, just faster and with millions of rows.

▼ Check yourself (Strang's lecture example): eliminate $x + 2y + z = 2$, $3x + 8y + z = 12$, $4y + z = 2$.

$\ell_{21} = 3$: row 2 becomes $2y - 2z = 6$. $\ell_{31} = 0$: nothing to do. $\ell_{32} = 4/2 = 2$: row 3 becomes $5z = -10$. Pivots $1, 2, 5$. Back substitution: $z = -2$, then $2y + 4 = 6$ so $y = 1$, then $x + 2 - 2 = 2$ so $x = 2$.

5. When elimination breaks down

Pivots can never be 0: we divide by them to find multipliers. So what if a 0 lands in a pivot position? Strang separates two cases.

Temporary failure: fixed by a row exchange

METHOD · ROW EXCHANGE (PIVOTING)

When a pivot position holds 0, swap that row with a lower row that has a non-zero entry in the same column. Strang swaps only when forced. Software usually also swaps to bring the **largest** entry into the pivot position (**partial pivoting**), which keeps rounding errors small.

Replace the bond fund with a **credit fund** that also earns 12% but has no market exposure (beta 0). Its column is $(1, 12, 0)$. With target $(10, 126, 7.4)$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 12 & 15 & 126 \\ 1 & 0 & 1.2 & 7.4 \end{array}\right] \xrightarrow{R_2 - 12R_1,\ R_3 - R_1} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \mathbf{0} & 3 & 6 \\ 0 & -1 & 0.2 & -2.6 \end{array}\right]$$

A 0 in the second pivot position: Nifty and the credit fund earn the same, so removing x from the return rule removed y too. But there is a non-zero entry **below** it, so **exchange rows 2 and 3**:

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-1} & 0.2 & -2.6 \\ 0 & 0 & \boxed{3} & 6 \end{array}\right]$$

Now it is upper triangular, so back-substitute from the bottom row up:

1. **Row 3:** $3z = 6$, so $z = 2$.
2. **Row 2:** $-y + 0.2z = -2.6$. Put in $z = 2$: $-y + 0.4 = -2.6$, so $-y = -3$ and $y = 3$.
3. **Row 1:** $x + y + z = 10$. Put in $y = 3$, $z = 2$: $x + 5 = 10$, so $x = 5$.

₹5 lakh Nifty, ₹3 lakh credit fund, ₹2 lakh midcap. Check in the original rules: budget $5 + 3 + 2 = 10$ ✓; return $12(5) + 12(3) + 15(2) = 60 + 36 + 30 = 126$ ✓; exposure $5 + 0 + 1.2(2) = 7.4$ ✓.

Three pivots $(1, -1, 3)$ after the exchange, so the matrix is fine (invertible). Swapping the order of two equations never changes the solution.

Permanent failure: the matrix is singular

Now replace the midcap fund with the **balanced fund** from Lesson 1 (half Nifty, half bonds), column $(1, 9.5, 0.5)$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 9.5 & 111 \\ 1 & 0 & 0.5 & 7.4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-5} & -2.5 & -9 \\ 0 & -1 & -0.5 & -2.6 \end{array}\right] \xrightarrow{R_3 - 0.2R_2} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & \\ 0 & \boxed{-5} & -2.5 & \\ 0 & 0 & \mathbf{0} & \end{array}\right.$$

A 0 in the third pivot position and **nothing below it** to swap in. Only **two pivots**. The last row reads $0 = -0.8$: **no solution**. (With a target on the funds' plane, such as $(10, 95, 5)$, the last row becomes $0 = 0$ and there are infinitely many solutions.)

> **PIVOTS DECIDE EVERYTHING**
>
> An $n \times n$ matrix is **invertible** exactly when elimination (with row exchanges where needed) produces n **non-zero pivots**. Fewer pivots means **singular**: no solution or infinitely many, just as the pictures in Lesson 1 showed.

The 2×2 version: two Nifty funds give $\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, one pivot, and the return rule becomes $0 = -15$.

▼ Check yourself: in Strang's example, which number in place of the 8 causes a temporary failure, and which number in place of the last 1 causes a permanent one?

An $8 \to 6$ makes the second pivot 0, but row 3 has a 4 below it, so a row exchange fixes it. A last entry of -4 instead of 1 makes the third pivot $-4 - (-4) = 0$ with nothing below: permanent failure, singular matrix.

6. Elimination as matrices

> **METHOD · ELIMINATION MATRICES E_{IJ}**
> Write each row operation as a left multiplication by an elementary matrix: the identity with $-\ell_{ij}$ in position (i, j). All of elimination becomes $E_{32}E_{31}E_{21}A = U$.

In topic 3 we did three row operations by hand. Strang's main point is that **each of them is a multiplication by a matrix.** Once that is true, the whole of elimination fits in one line, and that line is what the next lessons build on.

How a matrix can change rows

Lesson 1's key fact was about columns: **matrix × column** mixes the columns (Lesson 1). Its twin is about rows:

> **ROW × MATRIX MIXES THE ROWS**
> $\begin{bmatrix} a & b & c \end{bmatrix} A = a\,(\text{row 1 of } A) + b\,(\text{row 2}) + c\,(\text{row 3}).$

Try it on our portfolio matrix with the row $\begin{bmatrix} -12 & 1 & 0 \end{bmatrix}$:

$$\begin{bmatrix} -12 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} = -12\,(1,1,1) + 1\,(12,7,15) = (0, -5, 3).$$

That is exactly the new return rule from step (2, 1): "return − 12 × budget". So a matrix whose rows are recipes like this can rebuild every row at once:

Row of the matrix	Recipe it follows	New row
$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$	keep row 1	budget, unchanged
$\begin{bmatrix} -12 & 1 & 0 \end{bmatrix}$	row 2 − 12 × row 1	return with x removed
$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$	keep row 3	exposure, unchanged

Stacking those three rows gives the **elimination matrix** E_{21}. It is the identity matrix I (which keeps every row) with one extra entry, -12, in row 2, column 1.

> **ELIMINATION MATRIX E_{IJ}**
> The identity matrix with $-\ell_{ij}$ in row i, column j. Multiplying by it subtracts ℓ_{ij} × row j from row i and leaves every other row alone.

The whole of topic 3, as three multiplications

Step (2, 1), $\ell_{21} = 12$: subtract 12 × budget from return.

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}} \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 1 & 0 & 1.2 \end{bmatrix}$$

Step (3, 1), $\ell_{31} = 1$: subtract budget from exposure.

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}}_{E_{31}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 1 & 0 & 1.2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & -1 & 0.2 \end{bmatrix}$$

Step (3, 2), $\ell_{32} = 0.2$: subtract 0.2 × the new row 2 from row 3.

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -0.2 & 1 \end{bmatrix}}_{E_{32}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & -1 & 0.2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix} = U$$

The same three matrices turn $b = (10, 111, 7.4)$ into $(10, -9, 7.4)$, then $(10, -9, -2.6)$, then $c = (10, -9, -0.8)$. Written as one line:

$$E_{32}\, E_{31}\, E_{21}\, A = U \qquad \text{and} \qquad E_{32}\, E_{31}\, E_{21}\, b = c.$$

One matrix for all of elimination

Matrix multiplication is **associative**: you can move the parentheses, $E_{32}(E_{31}(E_{21}A)) = (E_{32}E_{31}E_{21})A$. So multiply the three E's first and you get a single matrix that does every step at once:

$$E = E_{32}E_{31}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 1.4 & -0.2 & 1 \end{bmatrix}, \qquad EA = U.$$

Check its bottom row with the row rule: $1.4(1,1,1) - 0.2(12,7,15) + 1(1,0,1.2) = (1.4 - 2.4 + 1,\ 1.4 - 1.4 + 0,\ 1.4 - 3 + 1.2) = (0, 0, -0.4)$ ✓, the last row of U.

WATCH OUT: ORDER MATTERS

You may move parentheses, but you may **not** reorder: in general $AB \neq BA$. The first step sits next to A, the last step on the far left. Swap two of them and you get a different matrix:

$$E_{32}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 2.4 & -0.2 & 1 \end{bmatrix} \qquad \text{but} \qquad E_{21}E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 0 & -0.2 & 1 \end{bmatrix}.$$

Why: $E_{21}E_{32}$ does E_{32} *first*, so it subtracts $0.2\times$ the **old** return row $(12, 7, 15)$ instead of the new one $(0, -5, 3)$. Same operations, different order, different result.

Notice the messy 1.4 in E, even though the multipliers were the clean numbers $12, 1, 0.2$. Going the other way, from U back to A, keeps them clean: that is **Lesson 4**, $A = LU$.

▼ **Check yourself: what does the row $\begin{bmatrix} 1 & 2 & 7 \end{bmatrix}$ times a 3 × 3 matrix produce? (Strang's example)**

A row: $1\times$ row 1 $+2\times$ row 2 $+7\times$ row 3 of that matrix.

7. Row exchanges: permutation matrices

METHOD · PERMUTATION MATRIX P

The identity with its rows reordered. PA reorders the rows of A; AP reorders its columns.

The row exchange in topic 5 (the credit fund) is also a matrix. After column 1 we had a 0 in the second pivot position:

$$M = \begin{bmatrix} 1 & 1 & 1 \\ 0 & \mathbf{0} & 3 \\ 0 & -1 & 0.2 \end{bmatrix}$$

To build the matrix that swaps rows 2 and 3, do that swap to the identity matrix:

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad P_{23}M = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \\ 0 & 0 & 3 \end{bmatrix}.$$

Read its rows as recipes: "take row 1", "take row 3", "take row 2". A matrix that only reorders rows is a permutation matrix. For the credit-fund system the whole elimination is $P_{23}\,E_{31}\,E_{21}\,A = U$.

Left acts on rows, right acts on columns

Take the two-fund matrix $A = \left[\begin{smallmatrix} 1 & 1 \\ 12 & 7 \end{smallmatrix}\right]$ and the swap $P = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$:

$$PA = \begin{bmatrix} 12 & 7 \\ 1 & 1 \end{bmatrix} \quad \text{(rows swapped)} \qquad\qquad AP = \begin{bmatrix} 1 & 1 \\ 7 & 12 \end{bmatrix} \quad \text{(columns swapped)}$$

> **THE RULE**
>
> A matrix on the **left** acts on rows: all of elimination works this way. A matrix on the **right** acts on columns. $PA \neq AP$: one more case of $AB \neq BA$.

In portfolio terms: PA lists the **rules** in a different order (return first, then budget), which never changes the answer. AP swaps the **funds**, so the first unknown becomes the bond amount: the answer comes out as $(3, 7)$ instead of $(7, 3)$.

8. Undoing a step: inverses

> **METHOD · INVERSE OF AN ELIMINATION STEP**
>
> Flip the sign of the multiplier: E_{ij}^{-1} adds back what E_{ij} subtracted. For a row exchange, swap again. Inverses of whole matrices use the **Gauss–Jordan method**, covered in Lesson 3.

E_{21} subtracted 12 × budget from return. To undo it, **add 12 × budget back**. That is the identity with $+12$ where E_{21} had -12:

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Check it on the portfolio: E_{21} turned the return row $(12, 7, 15)$ into $(0, -5, 3)$. Adding 12 × budget gives $(0, -5, 3) + 12(1, 1, 1) = (12, 7, 15)$: the original return rule is back. So $E_{21}^{-1}(E_{21}A) = A$, and $E_{21}^{-1}E_{21} = I$.

> **INVERSE**
>
> The matrix that undoes E is its inverse, written E^{-1}, with $E^{-1}E = I$. For an elimination matrix, just **flip the sign of the multiplier**. For a row exchange, **swap again**: $P_{23}^{-1} = P_{23}$.

WHY ELIMINATION GIVES THE RIGHT ANSWER

Every step can be undone, so no information is lost or added. Any portfolio that satisfies the original rules $Ax = b$ also satisfies the new rules $Ux = c$, and the other way round. That's why the answer we read off U, ₹5 lakh, ₹3 lakh and ₹2 lakh, is the answer to the original question.

Inverses of whole matrices (not just single steps), and when they exist, are the next lesson.

▼ **Check yourself: what is the inverse of E_{32}, which subtracts 0.2 × row 2 from row 3?**

Add 0.2 × row 2 back to row 3: the identity with $+0.2$ in row 3, column 2.